

Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

Lecture 2: Basic Statistics for Risk Management and Actuarial Science

Reference Handout (Article Version)

Department of Statistics and Data Science, The Chinese University of Hong Kong
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Link to Risk Management & Actuarial Science

What this handout is: A compact, inquiry-driven reference of **essential definitions, structural relationships, modelling gatekeepers, and Socratic prompts** for Statistics in risk management and actuarial science.

What it is not: A proof notebook. Whenever a proof would normally appear, we replace it by **decision points**—the logical gates where reasoning can fail or succeed, and where modelling choices become defensible.

Structural Core (★)

Depth markers (self-selected):

- ★ **Core:** Must-know definitions and relationships.
- ★★ **Insight:** Failure points and why the structure works.
- ★★★ **Extension:** Generalize, stress-test, and model consequences.

Pause-and-Think

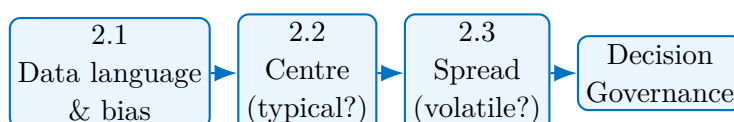
Name your dominant failure mode (pick one): **definition slips, units/context slips, interpretation slips.**

Then, for every page, ask: “Where would my failure mode appear here, and how would I catch it in 10 seconds?”

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Pipeline map (data → centre → spread → governance):

Classify data + check bias → choose a centre (mean/median/mode) → quantify spread (IQR/SD) → decide what the summary *allows you to claim* (pricing, reserving, capital planning).



Pause-and-Think

Single-point fragility test: Identify one place in this pipeline where a single wrong assumption can create a *five-year* premium error or a *capital planning* mistake. Name the assumption.

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Section 2.1 — Basic Statistics (Language Before Calculation)

Cognitive Trigger

A flood-risk insurer assesses a region using three kinds of observations collected over many years:

- Exact river water levels during typhoons (metres).
- Resident survey answers such as “Very High”, “Moderate”, “Negligible”.
- Annual counts of households making claims.

Non-routine questions:

1. Why is “add the survey answers and divide” not just wrong, but *meaningless*?
2. How can measurement rounding create long-horizon forecasting errors?
3. How can biased sampling distort expected losses *even if calculations are perfect*?

Structural Core (★)

Four statistical entities (language for what exists):

- **Population:** The entire group under study (the “whole reality”).
- **Sample:** A subset selected from the population (the “data you actually have”).
- **Parameter:** A numerical description of the population (unknown, fixed).
- **Statistic:** A numerical description computed from the sample (known, variable).

Key relation: A statistic is an **estimator** of a parameter.

Proof Decision Points (★★)

Gate 1 (What are you allowed to claim?): A sample statistic is *not* the population parameter. To speak responsibly you must decide:

- Are you estimating the population, or describing the sample only?
- If estimating: what is the sampling mechanism (random? biased?)?
- What is the error channel: sampling error vs bias vs measurement error?

Decision point: In risk work, the dangerous mistake is not “small error” but *unnoticed error*.

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If a sample mean loss is below the true population mean loss, reserves may be set too low. In risk systems, **small under-estimation** can become **solvency risk**.

Structural Core (★)

Notation map (population vs sample): Greek often marks population “truth”; Roman marks sample “estimate”.

Measure	Population	Sample
Size	N	n
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard deviation	σ	s
Proportion	P	p

Pause-and-Think

Give a one-sentence definition for each: population, sample, parameter, statistic.

Then answer: *Which one is “unknown but fixed”? Which one is “known but unstable”?*

2.1A — Types of Data (What mathematics is legal?)

Structural Core (★)

Qualitative data (categories):

- **Nominal:** No natural order (labels).
- **Ordinal:** An order exists, but gaps are not numerically meaningful.

Quantitative data (numbers):

- **Discrete:** Countable values.
- **Continuous:** Measurable values on a continuum.

Proof Decision Points (★★)

Gate 2 (Legitimate operations): The data type determines what algebra is meaningful.

- Nominal: Equality/inequality and counting; arithmetic averages are meaningless.
- Ordinal: Comparisons (greater/less) are meaningful; differences/ratios are not.
- Discrete numeric: Summation models are natural.
- Continuous numeric: Density/integration models are natural.

Decision point: When someone proposes an average, ask: *“Average of what structure?”*

Paired Examples: Works vs Fails

Works (meaning preserved):

- Claim counts per year → compute average count.
- Water level (metres) → compute average water level.

Fails (meaning destroyed):

- Policy type: {fire, flood, motor} → “mean policy type”.
- Survey: {Negligible, Moderate, Very High} → treat as numbers without a scale.

Repair strategies:

- For ordinal surveys: Use proportions in each category, or explicitly define a scoring rubric and justify it.

Pause-and-Think

Create your own counterexample: construct an ordinal survey where assigning scores 1–3 reverses the intended meaning.

Explain what went wrong: the scale, the spacing, or the interpretation?

2.1B — Discrete vs Continuous: two mathematical worlds

Structural Core (★)

Two worlds:

Discrete probability uses summation: $P(X \leq k) = \sum_{i=0}^k P(X = x_i).$

Continuous probability uses integration: $P(a \leq X \leq b) = \int_a^b f(x) dx.$

Deepening Layer (★★)

Why this matters in risk:

- Claim *counts* are discrete (you cannot have 3.7 claims) $\Rightarrow$ summation-based models.
- Water levels and loss *amounts* are often continuous $\Rightarrow$ density/integration-based models.

The danger is not the formula—it is choosing the wrong world, then computing perfectly inside the wrong world.

Pause-and-Think

Give a real example where data looks continuous but is actually discrete because of rounding or reporting.

How would that distort a probability estimate?

2.1C — Data collection and bias (Perfect arithmetic cannot fix biased inputs)

Structural Core (★)

Common data collection methods in insurance:

- **Census:** Observe every member of the population (expensive).
- **Sample survey:** Observe a subset (cheaper; introduces sampling error).
- **Administrative data:** Existing company databases (fast; may encode systematic bias).

Proof Decision Points (★★)

Gate 3 (Bias vs sampling error):

- **Sampling error** comes from randomness: different samples give different statistics.
- **Bias** comes from a distorted mechanism: the expected statistic is shifted.

Decision point: Ask “*Would the same type of household be missing every time?*” If yes, you are facing bias, not merely sampling variability.

Deepening Layer (★★)

Three common bias channels (risk language):

- **Selection bias:** Who gets included (e.g., only those who answer the phone).
- **Response bias:** What people report (exaggeration/under-reporting).
- **Measurement bias:** How you measure (e.g., rounding water levels systematically).

In risk governance, bias is a **model risk**: it can break pricing even if computation is flawless.

Google Gemini Exploration (Copy & Paste Prompt)

You are a risk analyst designing a flood-risk survey in Hong Kong. Propose 8 survey questions (mix of ordinal and numeric). For each question: (1) classify the data type (nominal/ordinal/discrete/continuous), (2) state which mathematical summaries are legitimate, (3) list one realistic bias channel (selection/response/measurement) and a concrete mitigation. Output as a table with columns: question | type | legal summaries | bias risk | mitigation.

Technology / Computation Box

Professional habit: After any computed statistic, write one sentence answering: “*What mechanism could bias this number, and how would I detect it quickly?*”

Section 2.2 — Measures of Central Tendency (Choosing a centre is a decision)

Cognitive Trigger

A claims dataset (HKD thousands) contains 20 property claims:

12.5, 15.2, 14.8, 16.1, 850, 15.5, 14.9, ...

All claims can share the same computed mean while the portfolio risk is completely different. **Non-routine questions:**

1. Which single number is *typical* for premium setting—and why can “typical” be a trap?
2. How can the wrong centre cause under-pricing or over-reserving?
3. If two portfolios have equal mean, what must you check next before claiming they are equally safe?

2.2A — The arithmetic mean (expected value language)

Structural Core (★)

Arithmetic mean (sample): for data $x_1, \dots, x_n$,

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}.$$

Population mean: μ .

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In actuarial language, mean connects to **expected value**. Many pricing principles begin with an expected loss term, then add risk loadings. But a mean can be fragile under heavy tails.

Proof Decision Points (★★)

Gate 4 (Mean balancing property):

The statement $\sum_{i=1}^n (x_i - \bar{x}) = 0$ is not a trick; it is forced by the definition of $\bar{x}$.

- **Decision point:** expand $\sum (x_i - \bar{x})$ into $\sum x_i - n\bar{x}$.
- **Decision point:** substitute the meaning of $\bar{x}$: $\sum x_i = n\bar{x}$.

Gatekeeper: If you change what “mean” means, the balancing property may fail. This is why the mean is the natural centre for squared-error objectives later.

Pause-and-Think

Try to invent a different “centre” c such that $\sum (x_i - c) = 0$ always holds.
What does your attempt force c to be? (You are discovering uniqueness.)

2.2B — Outlier sensitivity (centre vs tail)

Paired Examples: Works vs Fails

Works (mean reflects total mass):

If all claims are of similar magnitude, the mean is stable and interpretable for baseline pricing.

Fails (mean collapses under a single extreme):

One catastrophic claim can pull the mean far away from the typical claim level.

Repair strategies:

- Use median to describe typical behaviour.
- Use trimmed mean for robust summary.
- Model the tail separately (catastrophe layer) rather than forcing one number to do two jobs.

Deepening Layer (★★)

Why “one point can dominate” is structural: the mean is linear in the data:

$$\bar{x} = \frac{1}{n}(x_1 + \cdots + x_n).$$

A single very large x_j enters additively and is only divided by n . If x_j is enormous relative to the rest, division by n may not “dilute” it enough.

Pause-and-Think

Design a dataset of 20 values where 19 values are between 10 and 20, but the mean is above 50.
How large must the outlier be? Explain in one line of algebra.

2.2C — Median (robust centre)

Structural Core (★)

Median algorithm:

1. Sort data ascending.
2. If n is odd: take the middle value.
3. If n is even: average the two middle values.

Proof Decision Points (★★)

Gate 5 (Median decision points):

- The median depends on **order**, not size. This is why it resists outliers.

Gatekeeper: You must define a tie-breaking convention (especially for grouped data or repeated values).

Governance question: If a stakeholder disputes your “typical” value, can you defend your ordering and convention?

Self-Check (No working shown)

Find the median of the ordered dataset {12, 15, 18, 22, 25, 30}. **20.**

Pause-and-Think

Construct a dataset where the median is stable but the mean is unstable under a single added point. What does that say about which number is “typical”?

2.2D — Mode and skewness (quick tail detection)

Structural Core (★)

Mode: The most frequent value (or category).

Skewness diagnostic (rule of thumb):

- Symmetric: mean $\approx$ median $\approx$ mode.
- Right-skewed (common in losses): mean $>$ median.

Proof Decision Points (★★)

Gate 6 (Mode limits): The mode is not a tail tool; it ignores rare extremes.

Decision point: If your decision is about catastrophic solvency, a centre that ignores the tail cannot be sufficient.

Pause-and-Think

Explain how the mode of annual losses can be 0 while the mean is positive and large. What portfolio structure makes this happen?

2.2E — Grouped data (midpoint approximation as a modelling choice)

Structural Core (★)

Estimated mean for grouped data:

$$\bar{x} \approx \frac{\sum(f \cdot x_m)}{\sum f},$$

where f is frequency and x_m is class midpoint.

Proof Decision Points (★★)

Gate 7 (Grouped-data compromises):

- Midpoint approximation assumes values are roughly evenly distributed within each class.

Gatekeeper: Wide classes increase approximation error; narrow classes increase data collection cost.

Risk governance question: Would a regulator accept midpoint approximation for this decision, or demand raw data?

Pause-and-Think

Create two different datasets that produce the same grouped table (same class frequencies) but have very different true means.

What does that reveal about information loss in grouping?

2.2F — Weighted mean (exposure matters)

Structural Core (★)

Weighted mean:

$$\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}, \quad \text{and if } \sum w_i = 1, \quad \bar{x}_w = \sum w_i x_i.$$

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In insurance and finance, not every observation represents the same exposure. A weighted mean respects **economic importance**. Equal weighting can be a hidden fairness choice or a hidden modelling error.

Self-Check (No working shown)

Portfolio allocates 4 million at 6%, 5 million at 4%, 1 million at 25%. Weighted mean return: **6.9%**.

Proof Decision Points (★★)

Gate 8 (Weight representations):

- What do weights represent: money, exposure, credibility, or reliability?

Gatekeeper: If weights are arbitrary, the mean becomes arbitrary.

Modelling question: If an outlier has tiny exposure, should it influence the portfolio centre strongly?

Google Gemini Exploration (Copy & Paste Prompt)

Act as a senior actuary teaching a gifted student. Create three short scenarios where choosing the centre is a decision: (i) premium setting for ordinary claims with one catastrophe, (ii) fairness policy using household income with one billionaire, (iii) portfolio return with unequal allocations. For each scenario: propose mean/median/mode/trimmed mean/weighted mean candidates, state the gatekeeper assumptions for each, and give one numerical micro-example that makes one candidate fail. Do not solve; focus on decision points and failure modes.

2.2G — Technology as verification (not as authority)

Technology / Computation Box

R code (mean and trimmed mean):

```
claims <- c(12.5,15.2,14.8,16.1,850,15.5,14.9,
           15.3,15.0,14.7,15.1,14.6,15.4,15.2,
           15.3,16.0,14.8,15.1,15.5,14.9)
mean(claims)
mean(claims, trim = 0.05) # 5% trimmed mean
```

Interpretation habit: After each computed value, write one sentence linking it to a decision (pricing, reserving, fairness).

Google Gemini Exploration (Copy & Paste Prompt)

Generate a claims dataset of 30 values that includes: (a) a dense cluster around 15, (b) two moderate outliers around 60, and (c) one catastrophe above 600. Then: (1) compute mean, median, mode (if any), 10% trimmed mean, and a weighted mean where the catastrophe has weight 0.02 and ordinary claims share the remaining weight equally. (2) For each centre, write a one-sentence risk interpretation and a one-sentence failure mode. (3) Propose a governance rule: which centre is reported to customers, which to management, and which to regulators, with justification.

Section 2.3 — Measures of Dispersion (Centre alone is not risk)

Cognitive Trigger

Two portfolios share the same mean annual return of 7% over five years:

- Portfolio A: 5, 7, 9, 6, 8
- Portfolio B: -5, 20, -10, 35, -5

Non-routine questions:

1. Why can equal means hide fundamentally different liability risk?
2. If a minimum of 2% return is required every year, which portfolio is safer—and why?
3. What mathematical object quantifies “how violently outcomes swing”?

2.3A — Range (fast but fragile)

Structural Core (★)

Range:

$$\text{Range} = \max(x_i) - \min(x_i).$$

Proof Decision Points (★★)

Gate 9 (Range limits): It uses only two values. One unusual observation can dominate.

Decision point: Range is acceptable as a **quick alarm**, but rarely as a stand-alone risk metric.

Self-Check (No working shown)

Data: 10, 12, 12, 13, 15, 16, 20, 22, 95. Range = $95 - 10 = \mathbf{85}$.

Pause-and-Think

Construct two datasets with the same range but very different “typical spread”.
What does range fail to capture?

2.3B — Quartiles and IQR (robust spread)

Structural Core (★)

Quartiles:

- Q_1 : Value below which 25% of data lie.
- Median: Value below which 50% lie.
- Q_3 : Value below which 75% lie.

Interquartile Range (IQR):

$$\text{IQR} = Q_3 - Q_1.$$

Proof Decision Points (★★)

Gate 10 (Quartile conventions):

- You must choose a quartile convention (inclusive/exclusive methods differ in some cases).

Gatekeeper: IQR deliberately ignores extremes; if extremes are the risk of interest, IQR alone is insufficient.

Governance question: Are extremes “noise” or “signal” in this context?

Self-Check (No working shown)

Ordered data 10, 12, 12, 13, 15, 16, 20, 22, 95: $Q_1 = 12$, $Q_3 = 21$, IQR = **9**.

Deepening Layer (★★)

Why IQR resists outliers: It focuses on the central 50% of the data.

In operational risk reporting, this is often preferred because one exceptional case should trigger investigation, not redefine typical performance.

2.3C — Outliers via fences (rule-based anomaly detection)

Structural Core (★)

Box-plot fence rule:

$$\text{Lower fence} = Q_1 - 1.5 \text{ IQR}, \quad \text{Upper fence} = Q_3 + 1.5 \text{ IQR}.$$

Values outside the fences are flagged as outliers.

Proof Decision Points (★★)

Gate 11 (Outlier action triggers):

- An outlier is a **flag**, not a verdict.

Gatekeeper: Outliers may be data errors, rare-but-real events, or regime changes.

Risk governance: Decide whether to (i) correct, (ii) exclude with justification, (iii) model separately, or (iv) treat as a new normal.

Pause-and-Think

With $Q_1 = 12$, $Q_3 = 21$, IQR = 9, compute the upper fence.

Explain why 95 triggers a flag—and list two different actions depending on context.

2.3D — Variance and standard deviation (spread that “adds up”)

Cognitive Trigger

If you average ordinary deviations from the mean, you get zero:

$$\sum_{i=1}^n (x_i - \bar{x}) = 0.$$

So how can “average deviation” measure spread at all?

Proof Decision Points (★★)

Gate 12 (Squaring choice):

- Ordinary deviations cancel (+ and −).
- Squaring prevents cancellation and penalizes larger departures.

Decision point: Choose a loss function: squared deviations (variance), absolute deviations (MAD), or something else. Each choice encodes a risk attitude.

Structural Core (★)

Population vs sample formulas:

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}, \quad \sigma = \sqrt{\sigma^2},$$
$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}, \quad s = \sqrt{s^2}.$$

Deepening Layer (★★)

Why divide by $n - 1$ (Bessel correction) matters:

When estimating an unknown population variance from limited historical data, dividing by $n - 1$ corrects systematic underestimation that arises because $\bar{x}$ is itself estimated from the same sample. In risk terms: variance is a **second-order quantity**; small bias here can distort capital buffers.

Proof Decision Points (★★)

Gate 13 (Bessel correction as a decision):

- If you truly have the full population, divide by n .
- If you have a sample and want an unbiased estimator under standard assumptions, divide by $n - 1$.

Gatekeeper: You must decide whether you are in “population description” mode or “population estimation” mode.

Self-Check (No working shown)

Portfolio A: 5, 7, 9, 6, 8, mean = 7. Deviations: −2, 0, 2, −1, 1. Squared: 4, 0, 4, 1, 1. Sum = 10. Sample variance = $10/4 = 2.5$. Sample SD = $\sqrt{2.5} \approx 1.58$.

Technology / Computation Box

R code (Portfolio B comparison):

```
returns_B <- c(-5, 20, -10, 35, -5)
mean(returns_B)
var(returns_B)
sd(returns_B)
```

Interpretation habit: If two portfolios share the same mean, compute SD (or IQR) before calling either one “safer”.

Paired Examples: Works vs Fails

Works (complete story): Report both centre and spread (e.g., mean + SD, median + IQR).

Fails (partial story): Report the mean alone and claim “same mean implies same risk”.

Repair: Add dispersion + define a minimum-acceptable outcome constraint (liability threshold).

Pause-and-Think

If a fund requires at least 2% return every year, define a statistic that measures “how often we fail”. Is that a measure of centre, spread, or something else?

2.3E — Linear transformations (units and scaling governance)

Structural Core (★)

Linear transformations:

- If $y = x + b$, the standard deviation does *not* change.
- If $y = ax$, the standard deviation is multiplied by $|a|$.

Proof Decision Points (★★)

Gate 14 (Transformation limits):

- Adding a constant shifts the centre but not the spread.
- Scaling changes both centre and spread proportionally.

Gatekeeper: Never compare spreads across different units without normalization.

Pause-and-Think

Give a real example: currency conversion (USD to HKD).
Explain why both mean and SD scale, and why IQR scales too.

Google Gemini Exploration (Copy & Paste Prompt)

Create a short "risk reporting memo" comparing Portfolio A and Portfolio B. Include: (1) centre (mean) and at least two spread measures (SD and IQR or range), (2) a minimum-return constraint analysis: how many years fall below 2%, (3) a governance recommendation: which portfolio is acceptable for a guaranteed-liability product and why. Then add a section titled "Failure modes" listing 5 ways a reader could be misled if only one statistic is shown. Use precise language and avoid hand-waving.

Closing Synthesis

Structural Core (★)

Professional storyline (mirrors real risk work):

1. **Define the data and check bias** (classification + collection).
2. **Choose a centre** (what does “typical” mean for this decision?).
3. **Quantify spread** (how volatile is the outcome around that centre?).
4. **Translate into governance** (what claim can you responsibly make?).

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A complete description of risk needs both a **centre** and a **spread**. Correct definitions, correct centre, and correct spread form the backbone of pricing, reserving, and communication to stakeholders.

Pause-and-Think

Final challenge: Write a two-sentence rule for yourself:

- Sentence 1: What you must check before computing any statistic.
- Sentence 2: What you must report so that your summary cannot be accused of being misleading.